

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

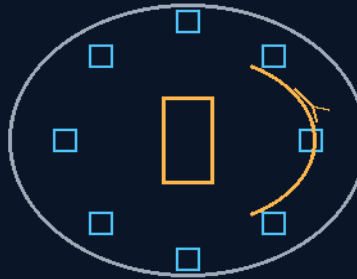
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 149

THE QUINN CENTRIFUGAL CALIBRATION MASS SCALE TURNTABLES

Spin to the slug's Earth weight, freeze the RPM
the un-fuckupable 1G window



steel, not software

EDGE CELLS · MASTER SLUG · RPM LOCK

CALIBRATE AGAINST REALITY

Metrology system — v1.0 in force

GP-IM-149

Revision v1.0 — 23 August 2026

150 of 290

VETTED BATCH 19 — PHYSICS PASS

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 149

PHASE 149: THE QUINN CENTRIFUGAL CALIBRATION MASS SCALE TURNTABLES — STATUS: CURRENT — PHYSICS PASS. A scale that cannot drift, because it calibrates against a lump of steel. In microgravity a spring balance lies and an inertia algorithm rots; Phase 149 weighs by centrifuge: strain scales mounted vertically at the turntable's outer edge, a solid-steel master slug of verified Earth mass spun until the scale reads its exact Earth weight — that RPM is frozen as the 1G window, and every unknown mass is then read against it, software-free. The same platform slices into departments: galley mini-centrifuges with sealed silicone cups (no airborne flour), research-deck sample scales, and flush floor carousels in the exercise vaults for crew body-mass audits. The vet graded the centrifugal calibration principle sound — a clean green.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-149, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the scale law as seated (contents line, the three bodies — perimeter track, master-slug calibration, departmental modularity — verbatim) — §2 why spin beats springs (graded) — §3 the system on the turntable — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 149: **Centrifugal Calibration Mass Scale Turntables**. Rejects linear acceleration shaking devices and complex computerized inertia algorithms. Positions mechanical strain scales vertically along the outer edge of a balanced turntable deck, spinning the table until it matches a solid-steel master reference slug to establish a software-free 1G measuring baseline.'

THE MASTHEAD LINE (verbatim): 'Industrial Metrology Domain: Centripetal Gravity Regeneration &

Terrestrial Master Calibration Loops [extracted verbatim from the combined masthead line]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Microgravity Scale Invalidation via Solid-State Rotation [extracted verbatim from the combined masthead line]'

THE VERTICAL PERIMETER TRACK ALIGNMENT (VERBATIM)

* Rationale: Rejects linear acceleration shaking devices and complex computerized inertia algorithms to eliminate calibration drift and structural mechanical failures across long runs. * Edge-Mounted Measuring Cells: Positions mechanical and electronic strain scales vertically along the outer circumference face of a balanced, low-power turntable deck. * Outward Kinetic Force Vectors: Table rotation flings target masses outward against the vertical scale plates, translating centripetal acceleration into a clean, linear gravity vector.

THE UN-FUCKUPABLE MASTER SLUG CALIBRATION (VERBATIM)

* Physical Earth-Baseline Anchors: Utilizes solid-steel master reference weights of verified mass properties to lock down the system's operational velocity parameters prior to sampling. * Velocity RPM Lock Gates: Accelerates the turntable until the scale registers the exact Earth-weight value of the master reference slug, freezing the motor speed to establish a perfect 1G window. * Software-Free Mass Tracking: Replaces software calculations with direct mechanical physics; removing the test slug allows raw samples or crew members to be weighed with zero conversion errors.

MULTI-DEPARTMENT SCALE MODULARITY (VERBATIM)

* Phase 84 Galley Adaptations: Deploys countertop mini-centrifuge housings within cooking bays, using sealed silicone cups to measure recipes cleanly without airborne flour contamination. * Phase 12/60 Research Integration: Slices high-precision sample scales into exobiological and hydro-sampling decks to ensure rapid, un-adulterated material-mass auditing.

* Phase 100 Crew Health Auditing: Installs flush floor-mounted carousels inside exercise vaults, enabling personnel to check body-mass changes to secure biological stability profiles.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE PRINCIPLE, AFFIRMED: centripetal force is $m \cdot v^2 / r$ — a mass on a radius at a locked RPM produces a force exactly proportional to its mass, with no gravity required and no spring to age. Weighing by rotation is sound metrology.

THE MASTER-SLUG LOCK, AFFIRMED AS PRACTICE: calibrating the RPM against a verified physical mass — then freezing that speed — is the un-fuckupable move the phase names it: the reference is a lump of

steel, not a model. Software-free is not a slogan here; it is the elimination of an entire drift class.

THE HONEST BOUND: the window is exact at the slug's radius and clean for rigid masses; powders, fluids and live crew slosh and redistribute — hence the sealed cups and the carousel protocol. The phase already carries its own answers.

§3 — THE SYSTEM ON THE TURNTABLE

- THE TRACK: strain scales vertical at the turntable's outer edge — mechanical and electronic cells.
- THE SLUG: solid-steel master reference weights of verified mass — the Earth-baseline anchor.
- THE LOCK: spin until the scale reads the slug's exact Earth weight; freeze that RPM — the 1G window.
- THE GALLEY: countertop mini-centrifuge housings, sealed silicone cups (Phase 84).
- THE LAB: high-precision sample scales in the Phase 12/60 research decks.
- THE CREW: flush floor carousels in the exercise vaults (Phase 100) — body-mass auditing.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE MASTER-SLUG DOCTRINE (the phase's own name): calibrate against physical reality — a verified lump, not a calculation.
- THE SOFTWARE-FREE DOCTRINE: where a mechanical lock suffices, no code is written — drift has no address to live at.
- THE METROLOGY-EVERYWHERE PATTERN: one platform, four departments — the file's reuse instinct applied to weighing.
- THE VET'S CLEAN GREEN: one of the batch's unconditional passes — principle sound, no yellow arithmetic attached.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the turntable: rigid disc, vertical edge-mounted strain cells, verified master slugs.
- STEP 2 — Lock the window: spin to the slug's Earth-weight reading; freeze and log the RPM.
- STEP 3 — Prove the read: unknown masses against the window — linearity and repeatability logged.
- STEP 4 — Slice the departments: the galley cup rig, the lab sample scale, the crew carousel.

- STEP 5 — Publish the ledger: calibration curves and slug certificates — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 84 — the galley: the countertop mini-centrifuge adaptation.
- PHASE 12/60 — the research decks: sample-mass auditing.
- PHASE 100 — crew health: the exercise-vault carousels.
- PHASE 150 — the staging chutes (GP-IM-150): the metrology family — measurement hardware the file already owns.

§8 — DO-NOT-BUILD

- DO NOT weigh by spring in microgravity — there is nothing for a spring to fight.
- DO NOT calibrate from software — the slug is the reference; code drifts, steel doesn't.
- DO NOT weigh sloshables open — sealed cups and the carousel protocol exist for that.
- DO NOT move the window casually — the frozen RPM is the calibration; re-lock against the slug.

§9 — OWED

OWED: nothing flagged by the vet — a clean green. Ordinary bench duty stands per the series law: linearity and repeatability data from the build (§6), published with the chapter's ledger.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-149, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, second of the 20-chapter 'ok next 20, that will take us past gemini' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576 and 587): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master

record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 149